Yijia Kang, Manasvi Vardham
Professor Abhishek
NLP and Large Language Models
08 March 2026

RAG-Powered Financial Announcement Chatbot

System Architecture

This project builds a Retrieval-Augmented Generation (RAG) chatbot that answers questions about corporate financial announcements. Instead of relying only on a language model, the system first retrieves relevant documents from a database and then uses them as context to generate an answer. The system has four main components:

- Document processing
- Vector database
- Retrieval pipeline
- Conversational interface

Document Processing

The dataset contains over 1,800 financial announcements from the Saudi Stock Exchange. Each document includes the announcement title, date, summary, and financial impact. Because announcements can be long, each document is split into smaller chunks using a sliding window approach:

- Chunk size: 800 characters
- Overlap: 100 characters

Vector Database

The processed chunks are stored in ChromaDB, which acts as the system's vector database.

- Each chunk is converted into an embedding using the all-MiniLM-L6-v2 sentence transformer model.
- These embeddings capture the semantic meaning of the text.
- The database is persistent, so it does not need to be rebuilt every time the system runs.

Retrieval Pipeline

When a user asks a question, the system follows a two-stage retrieval process.First, the query is embedded and the vector database retrieves the top 20 most similar chunks. Then a cross-encoder re-ranking model evaluates each query-document pair and selects the top 5 most relevant chunks.This improves retrieval accuracy because the cross-encoder better understands how well each document actually answers the question. The selected chunks are then passed to the FLAN-T5 language model, which generates the final response using the retrieved context.

Conversation Memory and Interface

The chatbot also includes conversation memory, which stores the last five conversation turns. This allows the system to understand follow-up questions that refer to earlier messages.The system is deployed using Streamlit, which provides a simple web interface where users can:

- Ask questions about financial announcements
- Receive generated answers
- View source citations from the original documents

Challenges

One challenge was retrieval accuracy. Financial announcements often contain similar language, which can make it difficult for a simple vector search to return the most relevant results. To address this, the system uses a two-stage retrieval pipeline with cross-encoder re-ranking, which significantly improves the relevance of the retrieved documents.

Another challenge was handling multi-turn conversations. Follow-up questions often lack enough context by themselves. For example, a user may ask “When was it announced?” without repeating the company name. To solve this, the chatbot stores the previous conversation in memory and includes it in the prompt sent to the language model.

A third challenge was preventing the language model from generating unsupported information. Large language models sometimes produce confident answers that are not grounded in data. To reduce this risk, the system forces the model to generate answers using only the retrieved context and also displays source citations so users can verify the information.

Results

The final system successfully demonstrates how a RAG architecture can be used to build a domain-specific chatbot for financial information. The chatbot can:

- Retrieve relevant financial announcements from a large dataset
- Generate answers grounded in real documents
- Provide source citations for transparency
- Support multi-turn conversations using memory
- Run through a simple web interface

Testing showed that the two-stage retrieval process significantly improves answer quality, since the cross-encoder helps identify the most relevant documents before answer generation.